

Практика 3.1

В ходе работы были созданы два микросервиса:

- user-service (работа с пользователями)
- order-service (работа с заказами)

Для каждого микросервиса был создан Dockerfile.

Для запуска микросервисов использовался docker-compose.yml.

Микросервисы были успешно запущены с помощью команды:
docker compose up --build

После запуска была проверена работа сервисов через браузер:

GET /users — получение списка пользователей

GET /orders — получение списка заказов

GET /orders/1 — получение заказа с данными пользователя

Было проверено взаимодействие между микросервисами.

Структура корневой папки

```
C:\Users\Malika\Desktop\microservices-demo>dir
Том в устройстве C не имеет метки.
Серийный номер тома: 7A77-58A6

Содержимое папки C:\Users\Malika\Desktop\microservices-demo

09.04.2026  17:39    <DIR>          .
09.04.2026  17:39    <DIR>          ..
09.04.2026  16:57             390 docker-compose.yml
09.04.2026  17:43    <DIR>          order-service
09.04.2026  17:44    <DIR>          user-service
                1 файлов             390 байт
                4 папок   184 506 744 832 байт свободно

C:\Users\Malika\Desktop\microservices-demo>
```

Структура папки юзер

```

C:\Users\Malika\Desktop\microservices-demo\user-service>dir
Том в устройстве C не имеет метки.
Серийный номер тома: 7A77-58A6

Содержимое папки C:\Users\Malika\Desktop\microservices-demo\user-service

09.04.2026  17:44    <DIR>          .
09.04.2026  17:44    <DIR>          ..
09.04.2026  17:52             1 321 app.py
09.04.2026  16:47             175 Dockerfile
09.04.2026  16:42              12 requirements.txt
09.04.2026  16:44             161 users.json
                4 файлов              1 669 байт
                2 папок   184 505 982 976 байт свободно

C:\Users\Malika\Desktop\microservices-demo\user-service>_

```

Структура папки ордер

```

C:\Users\Malika\Desktop\microservices-demo\order-service>dir
Том в устройстве C не имеет метки.
Серийный номер тома: 7A77-58A6

Содержимое папки C:\Users\Malika\Desktop\microservices-demo\order-service

09.04.2026  17:43    <DIR>          .
09.04.2026  17:43    <DIR>          ..
09.04.2026  17:52             1 972 app.py
09.04.2026  16:51             175 Dockerfile
09.04.2026  16:49             170 orders.json
09.04.2026  16:48              30 requirements.txt
                4 файлов              2 347 байт
                2 папок   184 506 761 216 байт свободно

C:\Users\Malika\Desktop\microservices-demo\order-service>_

```

содержимое файлов

```

C:\Users\Malika\Desktop\microservices-demo\user-service>type Dockerfile
FROM python:3.11-alpine

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 3001

CMD ["python", "app.py"]
C:\Users\Malika\Desktop\microservices-demo\user-service>_

```

```
C:\Users\Malika\Desktop\microservices-demo\user-service>type requirements.txt
flask==3.0.3
C:\Users\Malika\Desktop\microservices-demo\user-service>_
```

```
C:\Users\Malika\Desktop\microservices-demo\order-service>type app.py
from flask import Flask, jsonify, request
import json
import os
import requests

app = Flask(__name__)
DATA_FILE = os.path.join(os.path.dirname(__file__), 'orders.json')
USER_SERVICE_URL = os.getenv('USER_SERVICE_URL', 'http://user-service:3001')

def read_orders():
    with open(DATA_FILE, 'r', encoding='utf-8') as f:
        return json.load(f)

def write_orders(orders):
    with open(DATA_FILE, 'w', encoding='utf-8') as f:
        json.dump(orders, f, ensure_ascii=False, indent=2)

@app.get('/orders')
def get_orders():
    return jsonify(read_orders())

@app.get('/orders/<int:order_id>')
def get_order_by_id(order_id):
    orders = read_orders()
    order = next((o for o in orders if o['id'] == order_id), None)

    if not order:
        return jsonify({'error': 'Order not found'}), 404

    try:
        response = requests.get(f'{USER_SERVICE_URL}/users/{order["user_id"]}', timeout=5)
        if response.status_code == 200:
            user_data = response.json()
        else:
            user_data = {'error': 'User not found in user-service'}
    except requests.RequestException:
        user_data = {'error': 'Cannot connect to user-service'}

    return jsonify({
        'order': order,
        'user': user_data
    })

@app.post('/orders')
def create_order():
    data = request.get_json(silent=True) or {}
```

```
C:\Users\Malika\Desktop\microservices-demo\order-service>type Dockerfile
FROM python:3.11-alpine

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 3002

CMD ["python", "app.py"]
C:\Users\Malika\Desktop\microservices-demo\order-service>
```

```

C:\Users\Malika\Desktop\microservices-demo\order-service>type Dockerfile
FROM python:3.11-alpine

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 3002

CMD ["python", "app.py"]
C:\Users\Malika\Desktop\microservices-demo\order-service>type orders.json
[
  {
    "id": 1,
    "user_id": 1,
    "item": "Laptop",
    "amount": 1
  },
  {
    "id": 2,
    "user_id": 2,
    "item": "Mouse",
    "amount": 2
  }
]
C:\Users\Malika\Desktop\microservices-demo\order-service>

```

```

C:\Users\Malika\Desktop\microservices-demo>type docker-compose.yml
services:
  user-service:
    build: ./user-service
    container_name: user-service
    ports:
      - "3001:3001"
    networks:
      - micro-net

  order-service:
    build: ./order-service
    container_name: order-service
    ports:
      - "3002:3002"
    networks:
      - micro-net
    depends_on:
      - user-service

networks:
  micro-net:
    driver: bridge
C:\Users\Malika\Desktop\microservices-demo>

```

Проверка GET /users — получение списка пользователей.

```

C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost:3001/users
[{"email":"alice@example.com","id":1,"name":"Alice"}, {"email":"bob@example.com","id":2,"name":"Bob"}]
C:\Users\Malika\Desktop\microservices-demo>

```

Проверка GET /users/1 — получение пользователя по ID.

```

C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost:3001/users/1
{"email":"alice@example.com","id":1,"name":"Alice"}

```

Проверка GET /orders — получение списка заказов.

```
C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost:3002/orders
[{"amount":1,"id":1,"item":"Laptop","user_id":1},{ "amount":2,"id":2,"item":"Mouse","user_id":2}]
```

Проверка GET /orders/1 — заказ содержит данные пользователя из user-service.

```
C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost:3002/orders/1
{"order":{"amount":1,"id":1,"item":"Laptop","user_id":1,"user":{"email":"alice@example.com","id":1,"name":"Alice"}}
```

Добавление нового пользователя (POST /users).

```
C:\Users\Malika\Desktop\microservices-demo>curl.exe -X POST http://localhost:3001/users -H "Content-Type: application/json" -d '{"name":"Charlie","email":"charlie@example.com","email":"charlie@example.com","id":3,"name":"Charlie"}
```

Проверка GET /users после добавления пользователя.

```
C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost:3001/users
[{"email":"alice@example.com","id":1,"name":"Alice"}, {"email":"bob@example.com","id":2,"name":"Bob"}, {"email":"charlie@example.com","id":3,"name":"Charlie"}]
```

Добавление нового заказа (POST /orders).

```
C:\Users\Malika\Desktop\microservices-demo>curl.exe -X POST http://localhost:3002/orders -H "Content-Type: application/json" -d '{"user_id":1,"item":"Keyboard","amount":1}'
{"amount":1,"id":3,"item":"Keyboard","user_id":1}
```

Проверка GET /orders после добавления заказа.

```
C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost:3002/orders
[{"amount":1,"id":1,"item":"Laptop","user_id":1},{ "amount":2,"id":2,"item":"Mouse","user_id":2},{ "amount":1,"id":3,"item":"Keyboard","user_id":1}]
```

Проверка GET /orders/{id} — получение нового заказа с данными пользователя.

```
C:\Users\Malika\Desktop\microservices-demo>curl.exe http://localhost:3002/orders/3
{"order":{"amount":1,"id":3,"item":"Keyboard","user_id":1,"user":{"email":"alice@example.com","id":1,"name":"Alice"}}
```

localhost:3001

localhost:3001/users

Автоформатировать ☐

```
[{"email":"alice@example.com","id":1,"name":"Alice"}, {"email":"bob@example.com","id":2,"name":"Bob"}]
```

localhost:3002

localhost:3002/orders

Автоформатировать ☐

```
[{"amount":1,"id":1,"item":"Laptop","user_id":1},{ "amount":2,"id":2,"item":"Mouse","user_id":2}]
```

localhost:3002

localhost:3002/orders/1

Автоформатировать ☐

```
{"order":{"amount":1,"id":1,"item":"Laptop","user_id":1,"user":{"email":"alice@example.com","id":1,"name":"Alice"}}
```

